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HEAT DISSIPATION DEVICE HAVING IRREGULAR SHAPE

Abstract

A heat dissipation device includes a first casing and a second casing coupled to the first casing. The second casing includes a body having an inner surface and an outer surface opposite the inner surface, and a first portion and a second portion, each of the first and second portions having a different cross-sectional area. The heat dissipation device further includes a plurality of columns on the inner surface, and a first wick structure disposed on the inner surface and in the first portion and the second portion.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a divisional application of U.S. application Ser. No. 17/665,440, filed on Feb. 4, 2022, which is a continuation-in-part of U.S. Non-Provisional patent application Ser. No. 16/711,233 filed Dec. 11, 2019, which in turn claims priority under 35 U.S.C. § 119 to U.S. provisional patent application Nos. 62/783,717 filed Dec. 21, 2018, and 62/798,480 filed Jan. 30, 2019. This application also claims priority under 35 U.S.C. § 119 to Chinese patent application no. CN202120353801.1 filed Feb. 8, 2021, in the China National Intellectual Property Administration (CNIPA). The entire contents of all these applications are incorporated herein by reference.

BACKGROUND

[0002] Vapor chambers have higher efficiency in heat dissipation than heat pipes. A vapor chamber includes a casing and a wick structure. The casing defines a chamber for accommodating cooling fluid. The wick structure is disposed in the chamber. The casing has an evaporation section for absorbing heat and a condensation section for dissipating heat. The cooling fluid is evaporated into gaseous state in the evaporation section, and then turns into liquid state in the condensation section and is carried back to the section area by the wick structure, thereby creating circulating cooling fluid.

[0003] Electronic products have become lighter, slimmer, and more compact, and, as a result, vapor chambers are required to have an irregular shape in order to not interfere or obstruct nearby electrical components. In such a case, portions of the vapor chamber are required to have reduced dimensions, and the cross-sectional area of the wick structure in these portions is reduced compared to its cross-sectional area in other portions of the vapor chamber. In a vertically orientated vapor chamber, when a heat source is in thermal contact with the upper portion of the vapor chamber, cooling fluid in the vapor chamber is required to circulate against gravity. However, the flow of the cooling fluid stagnates in the reduced dimension portions due to the smaller cross-sectional area of wick structure in these portions. Thus, the flow of the cooling liquid is restricted through the narrow area and circulation of the cooling fluid in the vapor chamber is inhibited, thereby affecting the operation of the vapor chamber.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] The present disclosure is best understood from the following detailed description when read with the accompanying figures. It is emphasized that, in accordance with the standard practice in the industry, various features are not drawn to scale and are used for illustration purposes only. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0005] FIG. 1 is an exploded view of a vapor chamber according to embodiments of the disclosure.

[0006] FIG. 2A is a plan view of a second casing of the heat dissipation device in FIG. 1, according to embodiments of the disclosure.

[0007] FIG. 2B illustrates the encircled region 2B of FIG. 2A in greater detail.

[0008] FIG. 2C illustrates the encircled region 2C of FIG. 2A in greater detail.

[0009] FIG. 2D is a plan view of the second casing, according to embodiments of the disclosure.

[0010] FIG. 3 illustrates a plan view of a second casing including two wick structures, according to embodiments of the disclosure.

[0011] FIG. 4 illustrates a plan view of a second casing including three wick structures, according to embodiments of the disclosure.

[0012] FIG. 5A illustrates a plan view of a second casing and a wick structure, according to embodiments of the disclosure.

[0013] FIG. 5B illustrates a plan view of the second casing of FIG. 5A including two wick structures, according to embodiments of the disclosure.

[0014] FIG. 6A illustrates a plan view of a second casing and a wick structure, according to some embodiments.

[0015] FIG. 6B illustrates the second casing of FIG. 6A including the wick structure of FIG. 6A and an additional wick structure, according to embodiments of the disclosure.

[0016] FIG. 7A illustrates a plan view of a second casing including two wick structures, according to embodiments of the disclosure.

[0017] FIG. 7B illustrates a plan view of the second casing of FIG. 7A including a single wick structure, according to embodiments of the disclosure.

[0018] FIG. 8A illustrates a plan view of a second casing including two wick structures, according to embodiments of the disclosure.

[0019] FIG. 8B illustrates a plan view of a second casing including a wick structure, according to some embodiments.

[0020] FIG. 8C illustrates a cross-sectional view of the second casing in FIG. 8B, according to embodiments of the disclosure.

[0021] FIG. 8D illustrates a plan view of the second casing including a single wick structure **180**, according to embodiments of the disclosure.

[0022] FIG. 9 illustrates a plan view of a second casing including a wick structure, according to embodiments of the disclosure.

[0023] FIG. 10A is a cross-sectional view of a bundled wick structure including multiple wick fibers arranged around a central wick fiber, according to embodiments of the disclosure.

[0024] FIG. 10B-10F illustrate different configurations of the wick structures of FIG. 10A.

[0025] FIG. 10G is a cross-sectional view of a bundled wick structure including multiple wick fibers, according to embodiments of the disclosure.

[0026] FIGS. 10H, 10J, and 10K illustrate different configurations of the wick structures of FIG. 10G.

[0027] FIG. 11 illustrates a front view of a vapor chamber, according to embodiments of the disclosure.

[0028] FIG. 12 illustrates an exploded view of a vapor chamber in FIG. 11, according to embodiments of the disclosure.

[0029] FIG. 13 illustrates a plan view of a second casing including two wick structures, according to embodiments of the disclosure.

[0030] FIG. 14 illustrates a plan view of a first casing including a sheet-like wick structure, according to embodiments of the disclosure.

[0031] FIG. 15 illustrates a plan view of a second casing including two wick structures, according to embodiments of the disclosure.

[0032] FIG. 16 illustrates a plan view of a first casing including a sheet-like wick structure, according to embodiments of the disclosure.

[0033] FIG. **17** illustrates a plan view of a second casing including a wick structure, according to embodiments of the disclosure.

[0034] FIG. **18** illustrates a plan view of a second casing including a wick structure, according to embodiments of the disclosure.

DETAILED DESCRIPTION

[0035] It is to be understood that the following disclosure provides many different embodiments, or examples, for implementing different features of the disclosure. Specific embodiments or examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, dimensions of elements are not limited to the disclosed range or values, but may depend upon process conditions and/or desired properties of the device. Moreover, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed interposing the first and second features, such that the first and second features may not be in direct contact. Various features may be arbitrarily drawn in different scales for simplicity and clarity.

[0036] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly. In addition, the term “made of” may mean either “comprising” or “consisting of.”

[0037] FIG. **1** is an exploded view of a heat dissipation device **100** according to embodiments of the disclosure. FIG. **2A** is a plan view of a second casing **120** of the heat dissipation device **100** in FIG. **1**, according to embodiments of the disclosure. For the purposes of discussion herein, the heat dissipation device **100** is considered to be a vapor chamber. However, embodiments are not limited thereto and embodiments disclosed herein are equally applicable to other types of heat dissipation devices without departing from the scope of the disclosure.

[0038] Referring to FIGS. **1** and **2A**, the vapor chamber **100** is a generally flat, planar structure including a first casing **110**, a second casing **120**, a sheet-like wick structure **130**, and a wick structure **140**. The vapor chamber **100** is referred to as a thin heat spreader. The vapor chamber **100** has a thickness less than approximately 1 millimeter.

[0039] The first casing **110** and the second casing **120** may be composed of, for example, oxygen-free copper, silicon-containing copper alloy, aluminum-containing copper alloy, a combination thereof, and the like. Referring to FIG. **2A**, the second casing **120** includes a main body **121** and a plurality of supporting structures **122**. The supporting structures **122** protrude (or otherwise project) from an inner surface **123** of the main body **121** towards the first casing **110** and contact the first casing **110**. The supporting structures **122** reduce buckling of the first casing **110** and the second casing **120** and this limit deformation of the vapor chamber **100**. In some embodiments, the supporting structures **122** are column or pillar shaped having a diameter of about 0.4 mm to about 0.8 mm. However, embodiments are not limited in this regard. Other shapes and sizes are also possible provided the plurality of supporting structures **122** limit buckling of the first casing **110** and second casing **120**.

[0040] FIG. **2B** illustrates the encircled region **2B** of FIG. **2A** in greater detail. In some embodiments, and as illustrated, the supporting structures **122** are arranged in rows (or columns) with immediately adjacent rows offset from each other. For example, as illustrated in FIG. **2B**, the supporting structures **122** in row **131** are offset from the supporting structures **122** in row **133**. In some embodiments, the distance **DI** between alternate rows of supporting structures **122** (e.g., rows **131** and **135**) is twice the distance between adjacent rows of supporting structures **122** (e.g., rows

131 and rows **133**). In some embodiments, the distance **D1** is between 3.5 mm to 4.5 mm and the distance **D2** is between 1.5 mm to 2.5 mm. However, embodiments are not limited in this regard, and the distances **D1** and **D2** can be varied as required by application and design provided the plurality of supporting structures **122** limit buckling of the first casing **110** and second casing **120**. [0041] FIG. 2C illustrates the encircled region 2C of FIG. 2A in greater detail. As illustrated, the inner surface **123** includes a first fixing structure **175** along the outer periphery of the second casing **120** and a second fixing structure **177** surrounding the first fixing structure **175** and along the outer periphery of the second casing **120**. In some embodiments, the first fixing structure **175** and the second fixing structure **177** define a channel or groove **173** therebetween. The inner surface of the first casing **110** includes a corresponding protrusion that is received in the groove **173** for securing the first casing **110** and the second casing **120** to each other.

[0042] In some embodiments, the thickness of the first casing **110** is smaller than the thickness of the second casing **120**. As an example, the thickness of first casing **110** is about 0.1 mm to about 0.15 mm and the thickness of the second casing **120** is about 0.2 mm to about 0.35 mm. An overall thickness of the vapor chamber **100** including the first casing **110** and the second casing **120** is about 0.3 mm to 0.5 about mm. However, embodiments are not limited in this regard, and the thickness of the first casing **110** may be greater than the thickness of the second casing **120**.

[0043] The inner surface **123** is opposite to the outer surface **125** of the main body **121**. The outer surface **125** forms part of the outer surface of the vapor chamber **100** and is substantially planar. The first casing **110** is connected to the main body **121** of the second casing **120** by welding, soldering, brazing, or diffusion bonding. The second casing **120** has a first or “broad” portion **A1** and a second or “narrow” portion **A2**. The first portion **A1** has a width **W1** and the second portion **A2** has a width **W2**. The width **W1** is larger than the width **W2**. The first portion has length **L1** and the second portion has a length **L2**. The sum of the lengths **L1** and **L2** is the length **L3** of the longest side of the vapor chamber **100**. It will thus be understood that each of the first portion **A1** and the second portion **A2** has a different cross-sectional area. In an embodiment, the width **W1** of the first portion **A1** is approximately 80 millimeters (mm), the width **W2** of the second portion **A2** is approximately 18 mm. The length **L2** is about 90 mm to about 100 mm, and the length **L3** is about 120 mm to about 130 mm. However, embodiments are not limited in this regard and can be varied depending on the application and design requirements. In other embodiments, the width **W2** of the second portion **A2** may be equal to or less than half of the width **W1** of the first portion **A1**. In still other embodiments, the width **W2** of the second portion **A2** may be equal to or less than one-third of the width **W1** of the first portion **A1**.

[0044] The wick structure **130** is disposed between the first casing **110** and the second casing **120**. The wick structure **130** includes, for example, copper mesh. The wick structure **130** contacts the first casing **110** and the wick structure **140**. The wick structure **130** provides a flow path for the cooling fluid circulating in the vapor chamber **100**.

[0045] The wick structure **140** may be or include a bundle of copper wires twisted to form a single helical structure. The wick structure **140** is located in the gaps between the supporting structures **122** and contacts the inner surface **123** of the second casing **120**, and thereby is in contact with the second casing **120**. Although the wick structure **140** has been disclosed as a bundle of wires twisted into a helical structure, embodiments are not limited thereto. In other embodiments, the wick structure **140** may be or include a copper sintered powder wick structure including copper wires interlaced or twisted into a bundle. In some other embodiments, the wick structure **140** may be or include a plurality of the bundles of wires being interlaced or twisted. In still other embodiments, the wick structure **140** includes a screen mesh wick structure or groove wick structure.

[0046] In an embodiment and as illustrated, the wick structure **140** is longitudinally extending structure and relatively straight (without any curves or bends) disposed proximate upper ends of the first portion **A1** and second portion **A2**. One end **S** of the wick structure **140** is located in the second portion **A2** and proximate (but not contacting) a lateral edge **A21** of the second portion **A2**.

The lateral edge A21 is outer edge of the second portion A2, and thereby of the second casing 120. In some embodiments, the end S of the wick structure 140 may not contact with the edge A21 of the second portion A2, and may be spaced apart from the edge A21 of the second portion A2. The other longitudinally opposite end E of the wick structure 140 is located in the first portion A1. A heat source H is attached to the outer surface of the first casing 110 and thereby the outer surface of the vapor chamber 100 using a thermal paste, or other known techniques. FIG. 2 illustrates a vertical projection (dashed box) of the heat source H on the second casing 120. The heat source H is illustrated as square shaped for the sake of illustration. It will be understood that the shape of the heat source H is not limited to a square shape and the heat source H can have any desired shaped. The end E of the wick structure 140 is spaced (e.g., vertically, in FIG. 2) from the heat source H by a distance D. The distance D is approximately 18.5 mm, but embodiments are not limited thereto. In some embodiments, the end E of the wick structure 140 may overlap the heat source H or the end E and boundary (edges) of the heat source H may be coincident. It will be understood that the distance D between the end E of the wick structure 140 and the heat source H is not limited to any particular value and may be adjusted according to the size or shape of the vapor chamber and as per user and design requirements.

[0047] The vapor chamber 100 includes a working appendage 1150 that is cooperatively formed by a protrusion 1152 of the second casing 120 and a corresponding protrusion of the first casing 110. The working appendage 1150 includes a charging channel 1158 formed by the first casing 110 and the second casing 120. The charging channel 1158 fluidly communicates with the interior cavity of the vapor chamber 100 formed by the inner surface 123 of the second casing 120 and the inner surface of the first casing 110. The working appendage 1150 is used to fill the interior cavity of the vapor chamber 100 with working fluid and vacuuming out the air from the interior cavity. In some embodiments, the working appendage 1150 is centrally located along the width W1. However, in other embodiments, the working appendage 1150 is located offset from the central location or located on other sides or surfaces of the vapor chamber 100.

[0048] FIG. 2D is a plan view of the second casing 120, according to embodiments of the disclosure. In contrast to FIG. 2A, as illustrated, the wick structure 140 is located proximate a lower end of the second portion A2. The distance D between the end E of the wick structure 140 and the heat source H is thus reduced. The wick structure 140 is spaced from the lateral edge A21 of the second portion A2.

[0049] In some embodiments, the vapor chamber 100 may include more than one wick structure 140. FIG. 3 illustrates a plan view of a second casing 120a including two wick structures 141a and 142a, according to embodiments of the disclosure. The second casing 120a may be used instead of the second casing 120 in FIGS. 1 and 2 in the vapor chamber 100. The second casing 120a may be similar in some respects to the second casing 120 in FIG. 2, and therefore may be best understood with reference thereto where like numerals designate like components not described again in detail.

[0050] As illustrated, each of the wick structures 141a and 142a are longitudinally extending structures and are straight structures (without bends or curves) and are disposed on the second casing 120a. The wick structures 141a and 142a are located in the gaps between the supporting structures 122 and each contacts the inner surface 123 of the second casing 120a. In an embodiment, the wick structures 141a and 142a have different lengths. For example, and as illustrated, the wick structure 141a is longer than the wick structure 142a. The wick structures 141a and 142a contact each other along the longitudinal edges thereof and are arranged side by side. One end Sa1 of the wick structure 141a is located in the second portion A2 and is in contact with the edge A21 of the second portion A2. The longitudinally opposite end Ea1 of the wick structure 141a is located in the first portion A1 and spaced (e.g., vertically, in FIG. 3) from the heat source H. One end Sa2 of the wick structure 142a is located in the second portion A2, and the longitudinally opposite end Ea2 of the wick structure 142a is located in the first portion A1 and spaced (e.g., vertically, in FIG. 3) from the heat source H. As illustrated, the ends Ea1 and Ea2 are coincident

with each other (e.g., located at a same distance from the edge A21). However, in other embodiments, the ends Ea1 and Ea2 are non-coincident (e.g., located at different distances from edge A21) in the first portion A1.

[0051] Compared to the embodiment in FIG. 2, the second casing 120a in FIG. 3 includes the additional wick structure 142a extending from the second portion A2 toward the heat source H. The wick structure 142a improves circulation of cooling fluid in the vapor chamber 100, for example, against gravity. Thus, the efficiency of the vapor chamber 100 in dissipating heat is improved.

[0052] FIG. 4 illustrates a plan view of a second casing 120b including three wick structures 141a, 142a, and 143a, according to embodiments of the disclosure. The second casing 120b may be used instead of the second casing 120 in FIGS. 1 and 2 in the vapor chamber 100. The second casing 120b may be similar in some respects to the second casings 120 and 120a in FIGS. 2 and 3, and therefore may be best understood with reference thereto where like numerals designate like components not described again in detail.

[0053] Referring to FIG. 4, each of the wick structures 141a, 142a, and 143a are longitudinally extending structures and are straight structures (without bends or curves) and disposed on the second casing 120b. The wick structures 141a, 142a, and 143a are located in the gaps between the supporting structures 122 and each contacts the inner surface 123 of the second casing 120b. In an embodiment, the wick structures 141a, 142a, and 143a have different lengths. For example, and as illustrated, the wick structure 141a is longer than the wick structures 142a and 143a. The wick structures 141a and 142a contact each other along the longitudinal edges thereof and arranged side by side. The wick structure 143a is spaced (e.g., vertically, in FIG. 4) from the wick structures 141a and 142a. One end Sa1 of the wick structure 141a is located in the second portion A2 and contacts the edge A21 of the second portion A2. The longitudinally opposite end Ea1 of the wick structure 141a is located in the first portion A1 and spaced from the heat source H (illustrated in phantom). One end Sa2 of the wick structure 142a is located in the second portion A2, and the longitudinally opposite end Ea2 of the wick structure 142a is located in the first portion A1 and spaced from the heat source H. As illustrated, the ends Ea1 and Ea2 are coincident with each other (e.g., located at a same distance from the edge A21). However, in other embodiments, the ends Ea1 and Ea2 are non-coincident. One end Ea3 of the wick structure 143a is collinear (e.g., aligned) with the heat source H on the outer surface of first casing 110, and the other end Sa3 of the wick structure 143a is located in the first portion A1 and contacts a vertical edge A11 of the first portion A1. As illustrated, the wick structure 143a is entirely within the first portion A1. The horizontal distance between the end Ea3 of the wick structure 143a and the heat source H is less than the vertical distance between the end Ea2 of the wick structure 142a and the heat source H. Although, the wick structures 141a, 142a, and 143a are indicated as having different lengths, embodiments are not limited thereto. In some embodiments, the wick structures 142a and 143a may have the same lengths. In other embodiments, the end Sa1 of the wick structure 141a and end Sa2 of the wick structure 142a may be located at a same distance from the edge A21.

[0054] Compared to embodiment of FIG. 3, the second casing 120b in FIG. 4 includes an additional wick structure 143a extending from the edge of the first portion A1 toward the heat source H. The wick structure 143a further improves circulation of cooling fluid in the vapor chamber 100, for example, against gravity. Thus, the efficiency of the vapor chamber 100 in dissipating heat is further improved.

[0055] FIG. 5A illustrates a plan view of a second casing 120c and a wick structure 140c, according to embodiments of the disclosure. The second casing 120c may be used instead of the second casing 120 in FIGS. 1 and 2 in the vapor chamber 100. The second casing 120c may be similar in some respects to the second casing 120 in FIG. 2, and therefore may be best understood with reference thereto where like numerals designate like components not described again in detail.

[0056] As illustrated, the wick structure 140c is disposed in the second casing 120c and has at least one bend. The wick structure 140c is located in the gaps between the supporting structures 122 and

contacts the inner surface **123** of the second casing **120**. An end **Sc** of the wick structure **140c** is located in the second portion **A2** and contacts the edge **A21** of the second portion **A2** and the opposite end **Ec** of the wick structure **140c** is located in the first portion **A1** and overlaps at least part of the heat source **H** (illustrated in phantom) attached to the outer surface of the first casing **110**. In an example, and as illustrated the wick structure **140c** at least partially overlaps one side or end of the heat source **H** that is closer to the wick structure **140c**. For the sake of explanation, it is assumed that the heat source **H** has a generally rectangular shape including a first side **L1** and a second side **L2** opposite the first side **L1**, and a third side **L3** and a fourth side **LA** opposite the third side **L3**. The first side **L1** is closer to the edge **A21**. The fourth side **L4** is closer to the wick structure **140c**. The wick structure **140c** overlaps the fourth side **L4** of the heat source **H**.

[0057] As illustrated, the wick structure **140c** includes two sections **1401** and **1402**, having lengths **Y1** and **Y2**, respectively. The sections **1401** and **1402** are connected to each other at an angle (greater than 0° and less than 180°). The overall length of the wick structure **140c** is the sum of the lengths **Y1** and **Y2**. In an example, the overall length is approximately 118 mm, but the disclosure is not limited thereto. In other embodiments, the overall length of the wick structure **140c** increased or decreased as long as the wick structure **140c** is accommodated within the vapor chamber **100**. In an embodiment, an extent of the wick structure **140c** may be equal to the minimum distance **X** between the end **Sc** of the wick structure **140c** and the heat source **H**. More specifically, the minimum distance **X** is measured from the end **Sc** to a point on the heat source **H** nearest to the end **Sc** as projected vertically on the inner surface **123**. For instance, the distance **X** is approximately 105 mm. The heat dissipation efficiency of the vapor chamber including the wick structure **140c** is relatively higher when the overall length of the wick structure **140c** is equal to or less than about 126 mm.

[0058] The wick structure **140c** decreases a distance the cooling fluid has to circulate in the vapor chamber **100**. The length of the wick structure **140c** is not limited to any particular length and may be adjusted according to user and design requirements.

[0059] FIG. 5B illustrates a plan view of the second casing **120c** including the wick structure **140c** and a wick structure **150c**, according to embodiments of the disclosure. The wick structure **150c** is shaped and sized (or otherwise configured) similar to the wick structure **140c**. The wick structure **150c** contacts the wick structure **140c** at a bottom side thereof. As illustrated, the wick structure **150c** is located in the gaps between the supporting structures **122** and contacts the inner surface **123** of the second casing **120c**. An end **155** of the wick structure **150c** is located in the second portion **A2** and contacts the edge **A21** of the second portion **A2** and the opposite end **157** of the wick structure **150c** is located in the first portion **A1** and overlaps at least part of the heat source **H** (illustrated in phantom) attached to the outer surface of the first casing **110**. As depicted, the ends **155** and **157** of the wick structure **150c** are aligned with the ends **Sc** and **Ec** of the wick structure **140c**.

[0060] In some embodiments, the wick structure **150c** overlaps a same side of the heat source **H** as overlapped by the wick structure **140c**. In other embodiments, the wick structure **150c** overlaps a different side of the heat source **H**. The extent of the wick structure **150c** is the same as the extent of the wick structure **140c**.

[0061] The wick structure **150c** includes sections **1501** and **1502** having lengths **Y1** and **Y2**, respectively, and connected to each other at a same angle (greater than 0° and less than) 180° as the sections **1401** and **1402** of the wick structure **140c**. However, in other embodiments, the lengths of the sections **1501** and **1502** are different from the lengths of the sections **1401** and **1402**. In some embodiments, the length of the section **1501** is smaller than the length of the section **1401**. In some other embodiments, the length of the section **1502** is longer than the length of the section **1402**.

[0062] The location of the wick structure is not restricted to any particular location in the vapor chamber. FIG. 6A illustrates a plan view of a second casing **120d** and a wick structure **140d**, according to some embodiments. The second casing **120d** may be used instead of the second casing

120 of FIGS. **1** and **2** in the vapor chamber **100**. The second casing **120d** may be similar in some respects to the second casing **120c** in FIG. **5A**, and therefore may be best understood with reference thereto where like numerals designate like components not described again in detail.

[0063] Referring to FIG. **6A**, the wick structure **140d** has at least one bend and is disposed on the second casing **120d**. The wick structure **140d** is located in the gaps between the supporting structures **122** and contacts the inner surface **123** of the second casing **120d**. One end **Sd** of the wick structure **140d** is located in the second portion **A2**, and the other opposite end **Ed** of the wick structure **140d** is located in the first portion **A1** and overlaps the heat source **H** attached to the outer surface of the first casing **110**. Assuming, for the sake of explanation, that the heat source **H** is the same as the heat source **H** in FIG. **5A**, the wick structure **140d** overlaps the first side **L1**. As illustrated, the section having length **Y1** of the wick structure **140d** contacts a lower horizontal edge **A22** of the second portion **A2**.

[0064] The wick structure **140d** has a relatively shorter length than the wick structure **140c** of FIG. **5A**. The overall length of the wick structure **140d** is approximately 108 mm, but the disclosure is not limited thereto. In other embodiments, the overall length of the wick structure **140d** increased or decreased as long as the wick structure **140c** is accommodated within the vapor chamber **100**. An extent of the wick structure **140d** may be equal to the minimum distance **X** between the end **Sd** of the wick structure **140d** and the heat source **H**. More specifically, the minimum distance **X** is measured from the end **Sd** to a point on the heat source **H** nearest to the end **Sd** projected vertically on the inner surface **123**. For example, the distance **X** is approximately 100 mm. The heat dissipation efficiency of the vapor chamber including the wick structure **140c** is relatively higher when the overall length of the wick structure **140d** is equal to or less than about 120 mm.

[0065] FIG. **6B** illustrates the second casing **120d** including the wick structure **140d** and a wick structure **150d**, according to embodiments of the disclosure. The wick structure **150d** is shaped and sized (or otherwise configured) similar to the wick structure **140d**. The wick structure **150d** contacts the wick structure **140d** at the top side thereof. The wick structure **150d** is located in the gaps between the supporting structures **122** and contacts the inner surface **123** of the second casing **120d**.

[0066] An end **165** of the wick structure **150d** is located in the second portion **A2** and is spaced from the edge **A21** of the second portion **A2** and the opposite end **167** of the wick structure **150d** is located in the first portion **A1** and overlaps at least part of the heat source **H** (illustrated in phantom) attached to the outer surface of the first casing **110**. As depicted, the ends **165** and **167** of the wick structure **150d** are aligned with the ends **Sd** and **Ed** of the wick structure **140d**.

[0067] In some embodiments, the wick structure **150d** overlaps a same side of the heat source **H** as overlapped by the wick structure **140d**. In other embodiments, the wick structure **150d** overlaps a different side of the heat source **H**. The extent of the wick structure **150d** is the same as the extent of the wick structure **140d**.

[0068] The wick structure **150d** sections **1501** and **1502** having lengths **Y1** and **Y2**, respectively, and connected to each other at a same angle (greater than 0° and less than 180°) as the sections **1401** and **1402** of the wick structure **140d**. However, in other embodiments, the lengths of the sections **1501** and **1502** are different from the lengths of the sections **1401** and **1402**. In some embodiments, the length of the section **1501** is smaller than the length of the section **1401**. In some other embodiments, the length of the section **1502** is longer than the length of the section **1402**.

[0069] Referring to FIGS. **5A** and **6**, the wick structure **140d** in FIG. **6** is shorter than the wick structure **140c** in FIG. **5A**, and this reduces the path for the cooling liquid to circulate in the vapor chamber **100**. The length of the wick structure **140d** is not limited to any particular length and may be adjusted according to user and design requirements.

[0070] According to some embodiments, the shapes of the first casing and the second casing are not limited to any particular shape. FIG. **7A** illustrates a plan view of a second casing **120e** including two wick structures **141e** and **142e**, according to embodiments of the disclosure. The

second casing **120e** may be used in a vapor chamber that includes a first casing and a sheet-like wick structure similar to the first casing **110** and the sheet-like wick structure **130** in FIG. 1, respectively, but each having a shape corresponding to the shape of the second casing **120e**.

[0071] As illustrated in FIG. 7A, the second casing **120e** is generally H-shaped and has a first portion **A1**, a second portion **A2**, and a third portion **A3**. The third portion **A3** is located between the first and second sections **A1** and **A2**. The first portion **A1** has a width **W1**, the second portion **A2** has a width **W2**, and the third portion **A3** has a width **W3**. The width **W1** is equal to the width **W2**, and the width **W3** is smaller than the width **W1**. However, in other embodiments the widths **W1** and **W2** may be different, but each greater than the width **W3**. The wick structures **141e** and **142e** are disposed on the second casing **120e**. The wick structures **141e** and **142e** each have a generally L-shape and are located in the gaps between the supporting structures **122** and each contacts the inner surface **123** of the second casing **120e**.

[0072] The wick structure **141e** includes a first section **151** that is disposed in the first portion **A1**, second portion **A2**, and third portion **A3**. The wick structure **141e** includes a second section **152** disposed in the third portion **A3** and at an angle relative to the first section **151**. The first section **151** is connected to the second section **152** in the third portion **A3**. In an example, the first section **151** is perpendicular to the second section **152**. However, embodiments are not limited thereto, and the first section **151** is connected to the second section **152** at angles greater than or less than 90°.

[0073] The wick structure **142e** includes a first section **153** that is disposed in the first portion **A1**, second portion **A2**, and third portion **A3**. The wick structure **142e** includes a second section **154** disposed in the third portion **A3** and at an angle relative to the first section **153**. The first section **153** is connected to the second section **154** in the third portion **A3**. In an example, the first section **153** is perpendicular to the second section **154**. However, embodiments are not limited thereto, and the first section **153** is connected to the second section **154** at angles greater than or less than 90°. The first section **151** of the wick structure **141e** and the first section **153** of the wick structure **141e** contact each other.

[0074] The first section **151** includes an end **Ee1** of the wick structure **141e** in the first portion **A1**. The end **Ee1** overlaps the heat source **H** attached to the outer surface of the first casing **110**. The first section **153** includes an end **Ee2** of the wick structure **142e** in the first portion **A1**. The end **Ee2** overlaps the heat source **H** attached to the outer surface of the first casing **110**.

[0075] The second section **152** includes an end **Se1** of the wick structure **141e** in the third portion **A3**. The second section **154** includes an end **Se2** of the wick structure **142e** in the third portion **A3**.

[0076] As illustrated, the end **Se1** of the wick structure **141e** is located proximate (non-contacting) an upper edge **A22** of the second portion **A2** and the end **Se2** of the wick structure **142e** is located proximate a lower edge **A23** of the second portion **A2** opposite the upper edge **A22**. The second sections **152** and **154** are disposed proximate the edge **A21** of the second portion **A2**. The edge **A21** connects the upper edge **A22** and lower edge **A23**. In other embodiments, the end **Se1** may contact the upper edge **A22**, the end **Se2** may contact the lower edge **A23**, and the second sections **152** and **154** contact the edge **A21**.

[0077] Instead of two wick structures **141e** and **142e**, some embodiments include a single wick structure. FIG. 7B illustrates a plan view of the second casing **120e** including a single wick structure **170**, according to embodiments of the disclosure. The wick structure **170** includes a first section **171** disposed in the first portion **A1**, second portion **A2**, and third portion **A3**. The wick structure **170** includes a second section **172** disposed in the third portion **A3** and at an angle relative to the first section **171**. The first section **171** is connected to the second section **172** in the third portion **A3**. In an example, the first section **171** is perpendicular to the second section **172**. However, embodiments are not limited thereto, and the first section **171** is connected to the second section **172** at angles greater than 0° or less than 90°.

[0078] The first section **171** includes an end **Ee1** of the wick structure **170** in the first portion **A1**. The end **Ee1** overlaps the heat source **H** attached to the outer surface of the first casing **110**. The

second section **172** includes an end Se1 of the wick structure **170** in the third portion A3. The second section **172** includes an end Se2 opposite end Se1 located in the third portion A3. The second section **172** is disposed proximate the edge A21 of the second portion A2.

[0079] FIG. 8A illustrates a plan view of a second casing **120f** including two wick structures **141f** and **142f**, according to embodiments of the disclosure. The second casing **120f** may be used in a vapor chamber that includes a first casing and a sheet-like wick structure similar to the first casing **110** and the sheet-like wick structure **130** in FIG. 1, respectively, but each having a shape corresponding to the shape of the second casing **120f**.

[0080] The second casing **120f** includes a first portion A1 disposed between a second portion A2 and a third portion A3. The first portion A1 has a width W1, and the second portion A2 and the third portion A3 each have a width W2. The width W1 is larger than the width W2. In other embodiments, the second portion A2 and the third portion A3 may have different widths, but smaller than the width W1. The wick structures **141f** and **142f** are disposed on the second casing **120f**. The wick structures **141f** and **142f** each have a generally L-shape and are located in the gaps between the supporting structures **122** and each contacts the inner surface **123** of the second casing **120f**.

[0081] The wick structure **141f** includes a first section **161** that is disposed in the first portion A1. The wick structure **141f** includes a second section **162** disposed in the first portion A1 and the second portion A2 and at an angle relative to the first section **161**. The first section **161** is connected to the second section **162** in the first portion A1. In an example, the first section **161** is perpendicular to the second section **162**. However, embodiments are not limited thereto, and the first section **161** is connected to the second section **162** at angles greater than or less than 90°.

[0082] The wick structure **142f** includes a first section **163** that is disposed in the first portion A1. The wick structure **142f** includes a second section **164** disposed in the third portion A3 and at an angle relative to the first section **163**. The first section **163** is connected to the second section **164** in the first portion A1. In an example, the first section **163** is perpendicular to the second section **164**. However, embodiments are not limited thereto, and the first section **163** is connected to the second section **164** at angles greater than or less than 90°. The first section **161** of the wick structure **141f** and the first section **163** of the wick structure **142f** contact each other.

[0083] The first section **161** includes an end Ef1 of the wick structure **141f** located in the first portion A1. The end Ef1 overlaps the heat source H attached to the outer surface of the first casing **110**. The first section **163** includes an end Ef2 of the wick structure **142f** in the first portion A1. The end Ef2 overlaps the heat source H attached to the outer surface of the first casing **110**.

[0084] The second section **162** includes an end Sf1 of the wick structure **141f** in the second portion A2. The second section **164** includes an end Sf2 of the wick structure **142f** in the third portion A3.

[0085] As illustrated, the end Sf1 of the wick structure **141f** is located proximate an edge A32 of the second portion A2 and the end Sf2 of the wick structure **142f** is located proximate (non-contacting) an edge A33 of the third portion A3 opposite the edge A32. The second sections **162** and **164** are disposed proximate (but not contacting) an edge A31 of the second casing **120f** that forms (or otherwise defines) the first portion A1, the second portion A2, and the third portion A3 of the second casing **120f**. The edge A31 connects the edge A32 and edge A33. In other embodiments, the end Sf1 may contact the edge A32, the end Sf2 may contact the edge A33, and the second sections **162** and **164** contact the edge A31.

[0086] When the vapor chamber is used in a vertical manner (e.g., in a position wherein the heat source H is located at a level higher than the second portion A2 and third portion A3), the cooling fluid in the vapor chamber evaporates and turns to vapor while absorbing the heat generated by the heat source H. The cooling fluid (in vapor state) flows toward the second portion A2 and the third portion A3 and turns back to liquid state, and the cooling fluid can flow to the first portion A1 via the wick structures **141f** and **142f**. This configuration decreases a temperature difference between the first portion A1 and the second portion A2 and the third portion A3 by around 4 to 15 degrees

Celsius. The cooling fluid continuously circulates between the first portion A1 and the second portion A2 and the third portion A3, thereby dissipating the heat generated by the heat source H. [0087] FIG. 8B illustrates a plan view of a second casing 120g including a wick structure 142f, according to some embodiments. FIG. 8C illustrates a cross-sectional view of the second casing 120g taken along line 8B-8B in FIG. 8B. The second casing 120g may be similar in some respects to the second casing 120f in FIG. 8A, and therefore may be best understood with reference thereto where like numerals designate like components not described again in detail.

[0088] As illustrated in FIG. 8B, the second casing 120g includes a single wick structure 142f. The wick structure 141f is absent. The wick structure 142f is spaced from the edge A31. In some embodiments, the second section 164 is disposed mid-way in the third portion A3. In some embodiments, and as illustrated, the first section 163 of the wick structure 142f is centrally located in the first portion A1. However, embodiments are not limited in this regard and in some other embodiments, the first section 163 is located offset from the center of the first portion A1.

[0089] In some embodiments, the second casing 120g includes the wick structure 141f (FIG. 8A) and the wick structure 142f is absent. The wick structure 141f is spaced from the edge A31. In some embodiments, the second section 162 of the wick structure 141f is disposed mid-way in the second portion A2. In some embodiments, the first section 161 (FIG. 8A) of the wick structure 141f is centrally located in the first portion A1. However, in some other embodiments, the first section 161 is located offset from the center of the first portion A1.

[0090] Instead of two wick structures 141f and 142f, some embodiments include a single wick structure. FIG. 8D illustrates a plan view of the second casing 120f including a single wick structure 180, according to embodiments of the disclosure. The wick structure 180 includes a first section 181 disposed in the first portion A1. The wick structure 180 includes a second section 182 disposed in the first portion A1, second portion A2, and third portion A3 and at an angle relative to the first section 181. The first section 181 is connected to the second section 182 in the first portion A1. In an example, the first section 181 is perpendicular to the second section 182. However, embodiments are not limited thereto, and the first section 181 is connected to the second section 182 at angles greater than 0° or less than 90°.

[0091] The first section 181 includes an end Ef1 of the wick structure 180 in the first portion A1. The end Ef1 overlaps the heat source H attached to the outer surface of the first casing 110. The second section 182 includes an end Sf1 of the wick structure 180 in the second portion A2 and an end Sf2 opposite end Sf1 located in the third portion A3. The second section 182 is disposed proximate (but not contacting) the edge A31 that forms (or otherwise defines) the first portion A1, the second portion A2, and the third portion A3 of the second casing 120f. Although not illustrated, a working appendage similar to the working appendage 1150 in FIG. 2A is included in the second casing 120f in FIGS. 8A, 8B, and 8D. The working appendage is located at any desirable location along the sides of the second casing 120f.

[0092] The shape of the wick structure is not limited to any particular shape. In some embodiments, the wick structure may be in a zig-zag shape or any other desired shape.

[0093] FIG. 9 illustrates a plan view of a second casing 120h including a wick structure 141h, according to embodiments of the disclosure. The second casing 120h may be used in a vapor chamber that includes a first casing and a sheet-like wick structure similar to the first casing 110 and the sheet-like wick structure 130 in FIG. 1, respectively, but each having a shape corresponding to the shape of the second casing 120h.

[0094] As illustrated in FIG. 9, the second casing 120h is generally U-shaped and has a horizontally orientated first portion A1, and a second portion A2 and a third portion A3, each vertically oriented. The first portion A1 is located between the second and third sections A2 and A3. The first portion A1 has a width W1, the second portion A2 has a width W2, and the third portion A3 has a width W3. The width W2 is equal to the width W3, and the width W1 is less than the widths W2 and W3. However, in other embodiments the widths W2 and W3 may be different,

but each greater than the width W1. The wick structures **141h** is disposed in the second casing **120h** and in the gaps between the supporting structures **122**. The wick structures **141h** contacts the inner surface **123** of the second casing **120h**.

[0095] The wick structure **141h** includes a first section **191** horizontally oriented and disposed in the first portion A1, second portion A2, and third portion A3. The wick structure **141h** includes two vertically oriented sections, a second section **192** disposed in the second portion A3 and a third section **193** disposed in the third portion A3. Each of the first section **191**, second section **192**, and third section **193** are straight, longitudinal structures without any bends. The second section **192** and the third section **193** are angled relative to the first section **191**. The second section **192** and the third section **193** are connected to the first section **191** in the second portion A2 and the third portion A3, respectively. In an embodiment, and as illustrated, the second section **192** and the third section **193** are perpendicular to the first section **191**. However, embodiments are not limited in this regard, and the second section **192** and the third section **193** are connected to the first section **191** at angles greater than 0° or less than 90°.

[0096] The first section **191** includes an end Sg2 in the second portion A2, and proximate an outer vertical edge A22 of the second portion A2. The first section **191** includes an end Sg3 in the third portion A3 longitudinally opposite the end Sg2, and proximate an outer vertical edge A23 of the third portion A3. The first section **191** is located proximate a bottom edge A21 of the first portion A1, second portion A2, and third portion A3.

[0097] The second section **192** includes an end Eg2 in the second portion A2. The end Eg2 is longitudinally opposite the end of the second section **192** connected to the first section **191**. Similarly, the third section **193** includes an end Eg3 in the third portion A3. The end Eg3 is longitudinally opposite the end of the third section **193** connected to the first section **191**.

[0098] As illustrated, the end Eg2 overlaps the heat source H attached to the outer surface of the first casing **110** and over the second portion A2. However, in other embodiments, the end Eg3 overlaps the heat source H attached to the outer surface of the first casing **110** and over the third portion A3. A working appendage **1150** is located in the third portion A3. However in other embodiments, the working appendage **1150** is located in the first portion A1 or the second portion A2 along any desired edge.

[0099] FIG. 10A is a cross-sectional view of a bundled wick structure including multiple wick fibers **1010** arranged around a central wick fiber **1020**. In an embodiment, the wick fibers **1010** and **1020** include copper. However, the wick fibers **1010** and/or **1020** can include other materials that facilitate the flow of condensed fluid by capillary force through the wick structure formed using the wick fibers **1010** and **1020**.

[0100] FIGS. 10B-10F illustrate wick structures **140g**, **140h**, **140i**, **140j**, and **140k** including different arrangements of the wick fibers **1010** around the central wick fiber **1020**. In FIG. 10B, the wick structure **140g** includes the wick fibers **1010** twisted together in the shape of a helix to form a bundle (e.g., a structure similar to a braided rope). In FIG. 10C, the wick structure **140h** includes the wick fibers **1010** individually twisted and then twisted together in the shape of a helix to form a bundle. In FIG. 10D, the wick structure **140i** includes the wick fibers **1010** longitudinally extending and arranged side by side.

[0101] In FIGS. 10E and 10F, the wick structures **140j** and **140k** are formed by twisting the wick fibers **1010** in the shape of a helix to form a bundle. Due to the difference in the angle of twist of the wires in the wick structures **140j** and **140k**, the two wick structures **140j** and **140k** have a different tensile force.

[0102] In some other embodiments, the central wick fiber **1020** is absent in the wick structures **140g**, **140h**, **140i**, **140j**, and **140k**.

[0103] FIG. 10G is a cross-sectional view of a bundled wick structure including a plurality of wick fibers **1030** arranged in a circular manner. In an embodiment, the wick fibers **1030** include copper. However, the wick fibers **1030** can include other materials that facilitate the flow of condensed

fluid by capillary force through the wick structure formed using the wick fibers **1030**. FIG. **10H** illustrates a wick structure **140m** in which the plurality of wick fibers **1030** are loosely twisted together. FIG. **10J** illustrates a wick structure **140n** in which the plurality of wick fibers **1030** are longitudinally arranged. FIG. **10K** illustrates a wick structure **140p** in which the plurality of wick fibers **1030** are twisted together or in pairs to form a braided structure. In some other embodiments, the wick fibers **1010**, **1020**, and/or **1030** are flattened.

[0104] FIG. **11** illustrates a front perspective view of a vapor chamber **100i**, according to embodiments of the disclosure. FIG. **12** illustrates an exploded view of the vapor chamber **100i**, according to embodiments of the disclosure. FIG. **13** illustrates a plan view of a second casing **120i** of the vapor chamber **100i** in FIG. **11** including two wick structures, according to embodiments of the disclosure. FIG. **14** illustrates a plan view of a first casing **110i** of the vapor chamber **100i** in FIG. **11** including a sheet-like wick structure **130i**, according to embodiments of the disclosure. The vapor chamber **100i** also includes the working appendage **1150** (FIG. **2A**) cooperatively formed by the first casing **110i** and the second casing **120i**.

[0105] Referring to FIGS. **11** and **12**, the vapor chamber **100i** includes a first casing **110i** contacting a second casing **120i**. As illustrated in FIGS. **11-14**, the first casing **110i** includes a first inner surface **183** and a first outer surface **185**, and a mounting area **184** defined on the first outer surface **185**. The first inner surface **183** is opposite to the first outer surface **185**. The first outer surface **185** includes and forms part of the outer surface of the vapor chamber **100i** and is substantially planar except for the mounting area **184**. In some embodiments, the mounting area **184** is a raised (step) portion on the first outer surface **185**. A first heat source **H1** is thermally attached (or mounted) to the mounting area **184** and thereby to the outer surface of the vapor chamber **100i** using a thermal paste or other techniques. The heat source **H1** is not limited to any particular type of heat source **H1** and can include heat sources such as electronic circuits (CPUs). Similarly, the shape and size of the mounting area **184** are not limited to any particular shape and size, and the mounting area **184** can have any desired shape and size on which the first heat source **H1** can be correctly attached and from which the generated heat can be effectively dissipated. In addition, the location of the mounting area **184** is not limited to any particular location on the first outer surface **185**, and the mounting area **184** can be located anywhere on the first outer surface **185** as long as the first heat source **H1** can be correctly attached to the mounting area **184** and the first heat source **H1** can effectively dissipate heat.

[0106] As illustrated in FIGS. **12** and **14**, the sheet-like wick structure **130i** is a generally planar, wing-shape structure extending along a length direction (X-direction in FIG. **11**) of the vapor chamber **100i**. The sheet-like wick structure **130i** may be similar in some respects to the sheet-like wick structure **130** in FIG. **1**, and therefore may be understood with reference thereto where like components are not described again in detail. The sheet-like wick structure **130i** includes a first (or central) portion **85** including a first part **B1a** and a second part **B1b** adjacent and directly connected to the first part **B1a** in the length direction. The sheet-like wick structure **130i** further includes a second portion **B2** and a third portion **B3**, which form the opposite ends of the sheet-like wick structure **130i** in the length direction. The first part **B1a** and the second part **B1b** are located between the second portion **B2** and the third portion **B3**. As illustrated, the first part **B1a** is directly connected to the second portion **B2** in the length direction and the second part **B1b** is directly connected to the third portion **B3** in the length direction. Widths **W2** of the second portion **B2** and of the third portion **B3** are about the same (or within a certain range of each other (for example, $\pm 1-5$ mm)). Widths **W1** of the first part **B1a** of the second part **B1b** are about the same (or within a certain range of each other (for example, $\pm 1-5$ mm)). In some embodiments, the widths **W2** of the second portion **B2** and the third portion **B3** are less than the widths **W1** of the first part **B1a** and the second part **B1b**. In some other embodiments, the second portion **B2** and the third portion **B3** have different widths, and the first part **B1a** and second part **B1b** have different widths. However, even when the widths are different, the widths of the second portion **B2** and the third portion **B3**

are smaller than the widths of the first part **B1a** and second part **B1b**.

[0107] The thermal resistances of the second portion **B2** and the third portion **B3** are about the same (or within a certain value of each other), and the thermal resistances of the first part **B1a** and second part **B1b** are about the same (or within a certain value of each other). In some embodiments, the thermal resistances of the second portion **B2** and the third portion **B3** are greater than the thermal resistances of the first part **B1a** and second part **B1b**. In other embodiments, the second portion **B2** and the third portion **B3** have different thermal resistances, and the first part **B1a** and second part **B1b** have different thermal resistances. However, even when the thermal resistances are different, the thermal resistances of the second portion **B2** and the third portion **B3** are greater than the thermal resistances of the first part **B1a** and second part **B1b**. Thus, the heat transfer performance (or ability) of the first part **B1a** and second part **B1b** is higher than the heat transfer performance of the second portion **B2** and the third portion **B3**. In some embodiments, the thermal resistances of the second portion **B2** and the third portion **B3** are about the same (or within a certain value of each other), the thermal resistance of the first part **B1a** is less than the thermal resistances of the second portion **B2** and the third portion **B3**, and the thermal resistance of the second part **B1b** is less than the thermal resistance of the first part **B1a**. Additionally, in other embodiments, the thermal resistance of the second part **B1b** is greater than the thermal resistance of the mounting area **184**. Since the second part **B1b** is in thermal contact with the first heat source **H1** attached to the mounting area **184**, dissipation of heat generated from heat sources such as graphic processing units (GPUs) or central processing units (CPUs) is improved. It should be noted that, for the purposes of discussion herein, two parameters or quantities are considered to have a same value or a similar value when they have the same value or when a difference between their individual values is within an acceptable range, for example, $\pm 1-5$ units of measurement.

[0108] As illustrated in FIG. **13**, the second casing **120i** includes a second inner surface **283** and a second outer surface **285**, and a second mounting area **284**. The second inner surface **283** is opposite to the second outer surface **285**. The second outer surface **285** includes and forms at least some of the outer surface of the vapor chamber **100i** and is substantially planar. A second heat source **H2** is attached in the second mounting area **284** to the outer surface of the vapor chamber **100i** using a thermal paste, or other techniques. The second heat source **H2** is illustrated as square shaped for the sake of illustration. It will be understood that the second heat source **H2** is not limited to any particular shape and the second heat source **H2** can have any desired shape.

[0109] The second casing **120i** further includes an inner wick structure **140i**, illustrated as being arranged in the second casing **120i**. The inner wick structure **140i** is a generally planar, wing-shape structure extending along a length direction (X-direction in FIG. **13**) of the vapor chamber **100i**. The inner wick structure **140i** may be similar in some respects to the sheet-like wick structure **130i** in FIG. **12**, and therefore may be understood with reference thereto where like components are not described again in detail. The wick structure **140i** includes a first (or central) portion **95** including a first part **A1a** and a second part **A1b** adjacent to and directly connected to the first part **A1a** in the length direction. The wick structure **140i** further includes a second portion **A2** and a third portion **A3**, which form the opposite ends of the wick structure **140i** in the length direction. The first part **A1a** and second part **A1b** are disposed between the second portion **A2** and third portion **A3**. As illustrated, the first part **A1a** is directly connected to the second portion **A2** in the length direction and the second part **A1b** is directly connected to the third portion **A3** in the length direction. The first part **A1a** and second part **A1b** have a width **W1**, and the second portion **A2** and the third portion **A3** each have a width **W2**. Widths **W2** of the second portion **A2** and the third portion **A3** are about the same (or within a certain predetermined range/tolerance of each other (for example, $\pm 1-5$ mm)) and the widths **W2** of the first part **A1a** and second part **A1b** are about the same (or within a certain predetermined range/tolerance of each other (for example, $\pm 1-5$ mm)). The widths **W2** of the second portion **A2** and the third portion **A3** are less than the widths **W1** of the first part **A1a** and second part **A1b**. In other embodiments, the second portion **A2** and the third portion **A3** may

have different widths, and the first part **A1a** and second part **A1b** may have different widths. However, even when the widths are different, the widths **W2** of the second portion **A2** and the third portion **A3** are smaller than the widths **W1** of the first part **A1a** and second part **A1b**.

[0110] A second heat source **H2** is attached to the outer surface of the second casing **120i** and thereby the outer surface of the vapor chamber **100i** using a thermal paste, or other techniques. The second heat source **H2** is illustrated as square shaped for the sake of illustration. It will be understood that the shape of the second heat source **H2** is not limited to a square shape and the second heat source **H2** can have any desired shaped. In some embodiments, the thermal resistances of the second portion **A2** and the third portion **A3** are about the same (or within a certain value of each other, for example, ± 5 K/W of each other) and the thermal resistances of the first part **A1a** and second part **A1b** are about the same (or within a certain value of each other). In some embodiments, the thermal resistances of the second portion **A2** and the third portion **A3** are greater than the thermal resistances of the first part **A1a** and second part **A1b**. In other embodiments, the second portion **A2** and the third portion **A3** have different thermal resistances, and the first part **A1a** and second part **A1b** have different thermal resistances. However, even if the thermal resistances are different, the thermal resistances of the second portion **A2** and the third portion **A3** are greater than the thermal resistances of the first part **A1a** and second part **A1b**. Thus, the heat transfer performance (or ability) of the first part **A1a** and second part **A1b** is higher than the heat transfer performance (or ability) of the second portion **A2** and the third portion **A3**. In some embodiments, the thermal resistances of the second portion **A2** and the third portion **A3** are about the same (or within a certain value of each other), the thermal resistance of the first part **A1a** is greater than the thermal resistances of the second portion **A2** and the third portion **A3**, and the thermal resistance of the second part **A1b** is greater than the thermal resistance of the first part **A1a**. Thus, the second part **A1b** that is in thermal contact with the second heat source **H2** attached to the outer surface of the second casing **120i** provides an improved heat transfer performance of heat generated from heat sources such as graphic processing units (GPUs) or central processing units (CPUs).

[0111] In some embodiments, the heat transfer performance of the second part **B1b** in thermal contact with the first heat source **H1** attached to the outer surface of the first casing **110i** is higher than the heat transfer performance of the second part **A1b** in thermal contact with the second heat source **H2** attached to the outer surface of the second casing **120i**. In some embodiments, the heat transfer performance of the second part **B1b** in thermal contact with the first heat source **H1** attached to the outer surface of the mounting area **184** is about the same to the heat transfer performance of the second part **A1b** in thermal contact with the second heat source **H2** attached to the outer surface of the second casing **120i**.

[0112] In some embodiments, the sheet-like wick structure **130i** of the first casing **110i** and inner wick structure **140i** of the second casing **120i** are copper mesh wick structures, or other types of wick structures. In some embodiments, the thermal resistances of the first part **A1a**, second part **A1b**, second portion **A2**, and third portion **A3** are similar to the thermal resistances of the first part **B1a**, second part **B1b**, second portion **B2**, and third portion **B3**, respectively. In some embodiments, the thermal resistances of the first part **A1a**, second part **A1b**, second portion **A2**, and third portion **A3** are different than the thermal resistances of the first part **B1a**, second part **B1b**, second portion **B2**, and third portion **B3**, respectively. The thermal resistances can have any desired value as required by application or design provided the heat transfer performance of the second part **B1b** (that is in thermal contact with the first heat source **H1**) is higher than or equal to the heat transfer performance of the second part **A1b** (that is in thermal contact with the second heat source **H2**).

[0113] As illustrated in FIG. 12, the vapor chamber **100i** includes a plurality of supporting structures **122** for supporting the first casing **110i** and the second casing **120i** and reducing buckling of the casings **110i** and **120i**. As a result, deformation of the vapor chamber **100i** is reduced. In some embodiments, the supporting structures **122** are shaped as columns or pillars having a

diameter greater than 0.4 mm but less than 50 mm, and height (or thickness) greater than 0.1 mm but less than 30 mm. However, embodiments are not limited in this regard. Other shapes and sizes are also possible provided the plurality of supporting structures **122** provide the required support to limit buckling of the first casing **110i** and second casing **120i**. In some embodiments, a copper sintered powdered wick layer **124** surrounds some of the supporting structures **122**. In some embodiments, the copper sintered powdered wick layer **124** may include, for example, oxygen-free copper, silicon-containing copper alloy, aluminum-containing copper alloy, a combination thereof, and the like. In other embodiments, the copper sintered powdered wick layer **124** includes porous wick structures. In some embodiments, the copper sintered powdered wick layer **124** surrounds the supporting structures **122** located in the first part **A1a** and second part **A1b** and corresponding first part **B1a** and second part **B1b**. However, in other embodiments, the copper sintered powdered wick layer **124** surrounds all supporting structures **122**.

[0114] The supporting structures **122** surrounded by the copper sintered powdered wick layer **124** reduce the thermal resistance of the first part **A1a** and second part **A1b** having the second heat source **H2** attached to the outer surface of the second casing **120i** and corresponding first part **B1a** and second part **B1b** having the first heat source **H1** attached to the outer surface of the first casing **110i**. As a result, the heat transfer performance is improved. In some embodiments, only some of the supporting structures **122** located in the first part **A1a** and second part **A1b** and the first part **B1a** and second part **B1b** are surrounded with the copper sintered powdered wick layer **124**. In some embodiments, all the supporting structures **122** located in the first part **A1a** and second part **A1b** and corresponding first part **B1a** and second part **B1b** include the copper sintered powdered wick layer **124**. The supporting structures **122** having the copper sintered powdered wick layer **124** reduce the thermal resistance of the first part **A1a** and the second part **A1b** having the second heat source **H2** attached to the second casing **120i** and the first part **B1a** and the second part **B1b** having the first heat source **H1** attached to the first casing **110i**, and thereby the heat transfer performance is improved.

[0115] Referring to FIGS. **12** and **13**, illustrated are wick structures **141k** and **142k** disposed between the sheet-like wick structure **130i** and the inner wick structure **140i**, and each contacting contact the sheet-like wick structure **130i** and the inner wick structure **140i**. The wick structures **141k** and **142k** each have a generally serpentine shape (e.g., including multiple curves or turns) and are located in the gaps between the supporting structures **122**.

[0116] The wick structure **141k** includes a first section **111** that is disposed in the first part **A1a**. The wick structure **141k** includes a second section **112** disposed in the second portion **A2** and is inclined relative to the first section **111**. In some embodiments, the second section **112** is inclined at an angle greater than the ideal horizontal (0°) and less than the ideal vertical (90°). The first section **111** is connected to the second section **112** at the connection between the first part **A1a** and second portion **A2**. However, embodiments are not limited thereto, and the first section **111** can be inclined relative to the second section **112** at angles or degrees of curvatures greater than or less than 45° .

[0117] The wick structure **142k** includes a first section **113** that is disposed in the first part **A1a** and the second part **A1b**. The wick structure **142k** includes a second section **114** disposed in the third portion **A3** and is inclined relative to the first section **113**. In some embodiments, the second section **114** is inclined at an angle greater than the ideal horizontal (0°) and less than the ideal vertical (90°). The first section **113** is connected to the second section **114** at the connection between the second part **A1b** and third portion **A3**. However, embodiments are not limited thereto, and the second section **114** can be inclined relative to the second section **114** at angles or degrees of curvatures greater than or less than 45° . The first section **111** of the wick structure **141k** and the first section **113** of the wick structure **142k** do not contact each other.

[0118] In some embodiments, both wick structures **141k** and **142k** overlap the heat source **H2** located on the outer surface of the casing **120i**. In some embodiments, the wick structures **141k** and **142k** overlap the heat source **H2** via different sides (edges) of the heat source **H2**. In other

embodiments, the wick structure **142k** overlaps both heat sources **H1** and **H2**, and overlaps two or more sides (edges) of the heat source **H1**.

[0119] Referring to FIG. **13**, the first section **111** includes an end **E1** of the wick structure **141k**. The end **E1** overlaps the heat source **H2** attached to the first casing **110i**. The first section **113** includes an end **E2** of the wick structure **142k**. The end **E2** overlaps the heat source **H2** attached to the first casing **110i**. The wick structure **142k** overlaps the heat source **H1** attached to the outer surface of the second casing **120i**.

[0120] The second section **112** includes an end **S1** of the wick structure **141k**. The second section **114** includes an end **S2** of the wick structure **142k**. In an example, the first and second heat sources **H1**, **H2**, are attached to the outer surfaces of the mounting area **184** of the first casing **110k** and second casing **120k**, in first part **A1a** and the second part **A1b**, respectively. The thermal resistance of the second part **A1b** is lower than the thermal resistance of the first part **A1a**. The thermal resistance of the first part **A1a** is lower than the thermal resistances of the second portion **A2** and third portion **A3**. The heat transfer performance of the second part **B1b** in thermal contact with the first heat source **H1** attached to the outer surface of the mounting area **184** is higher than the heat transfer performance of the second part **A1b** in thermal contact with the second heat source **H2** attached to the outer surface of the second casing **120k**.

[0121] As illustrated, the end **S1** of the wick structure **141k** is located proximate (non-contacting) an edge **A32** of the second portion **A2** and the end **S2** of the wick structure **142k** is located proximate (non-contacting) an edge **A33** of the third portion **A3** opposite the edge **A32**. The second sections **112** and **114** are disposed proximate (non-contacting) an edge **A31** of the second casing **120i** that forms (or otherwise defines) the first part **A1a**, the second part **A1b**, the second portion **A2**, and the third portion **A3** of the second casing **120k**. Referring to FIG. **13**, the edge **A31** forms the lower edge of the second casing **120i** and connects the edge **A32** and edge **A33**. In other embodiments, the end **S1** may contact the edge **A32** and the end **S2** may contact the edge **A33**. In yet other embodiments, the second sections **112** and **114** contact the edge **A32**.

[0122] As illustrated, the end **E1** of the wick structure **141k** and the end **E2** of the wick structure **142k** are separated (non-contacting) from the edge **A31**. The end **E2** of the wick structure **142k** is located closer to the edge **A31** than the end **E1** of the wick structure **141k** and the first sections **111** and **113** (and thereby the ends **E1** and **E2**) are separated from each other either vertically (as illustrated) or horizontally. In other embodiments, the end **E2** of the wick structure **142k** is closer in proximity to the edge **A31** than the end **E1** of the wick structure **141k**. In some embodiments, the ends **E1** and **E2** cross each other in the vertical plane, for example, the **YZ** plane in FIG. **13** and the ends **E1** and **E2** are vertically separated from each other and overlapping in the horizontal plane, for example, the **XY** plane in FIG. **13**. However, in other embodiments, the ends **E1** and **E2** contact each other in the vertical plane, for example, the **YZ** plane in FIG. **13**, and, in the horizontal plane, for example, the **XY** plane, the ends **E1** and **E2** are either offset from each other or are aligned with each other. The first heat source **H1** attached to the outer surface of the mounting area **184** of the first casing **110i** is overlapped by the wick structure **142K**, and both the wick structure **141k** and the wick structure **142k** overlaps the second heat source **H2** attached to the outer surface of the second casing **120k**. In some embodiments, the heat source **H2** has a relatively higher power consumption than heat source **H1**. In some embodiments, the ends **E1** and **E2** of the wick structures **141k** and **142k** are located within the boundaries (edges) of the second heat source **H2**.

[0123] It will be understood that the location of the mounting area **184** may be varied depending on the application and design requirements, as long as the first heat source **H1** attached to the outer surface of the mounting area **184** of the first casing **110i** is overlapped by the wick structure **142k**.

[0124] The wick structures **141k** and **142k** decrease a distance the cooling fluid has to circulate in the vapor chamber. The length of the wick structures **141k** and **142k** are not limited to any particular length and may be adjusted according to user and design requirements.

[0125] When the vapor chamber is used in a vertical manner (e.g., in a position wherein the first

heat source H1 and/or second heat source H2 are (is) located at a level higher than the second portion A2 and third portion A3), the cooling fluid in the vapor chamber evaporates and turns to vapor while absorbing the heat generated by the first and/or second heat sources H1 and/or H2, respectively. The thermal resistance of the second part A1b is lower than the thermal resistance of the first part A1a. The thermal resistance of the first part A1a is lower than the thermal resistances of the second portion A2 and third portion A3. The heat transfer performance of the second part B1b in thermal contact with the first heat source H1 attached to the outer surface of the mounting area 184 is higher than the heat transfer performance of the second part A1b in thermal contact with the second heat source H2 attached to the outer surface of the second casing 120i.

[0126] In some embodiments, the thermal resistance of the second part B1b is higher than the thermal resistance of the mounting area 184. In some embodiments, the supporting structures 122 having the copper sintered powdered wick layer 124 surrounding the supporting structures 122 within the first part A1a and second part A1b, further reduces the thermal resistance of the first part A1a and second part A1b having the first heat source H1 attached to the outer surface of the mounting area 184 of the first casing 110k and second heat source H2 attached to the outer surface of the second casing 120i, further improving the heat transfer performance. The cooling fluid (in vapor state) flows toward the second portion A2 and the third portion A3 and turns back to liquid state, and the cooling fluid can flow to the first part A1a and second part A1b via the wick structures 141k and 142k. This configuration of the wick structures 141k and 142k, the portions of different thermal portions, and the copper sintered powered wick layers surrounding the supporting structures, decreases a temperature difference between the first part A1a and second part A1b and the second portion A2 and the third portion A3 by around 15 degrees Celsius or higher. The cooling fluid continuously circulates between the first part A1a and second part A1b and second portion A2 and third portion A3, thereby dissipating the heat generated by the first and/or second heat sources H1 and H2.

[0127] Compared to the embodiment of FIG. 8A, the wick structures 141k and 142k each have a generally flowing, sloping-ramped-shape and do not contact each other. Portions of the sheet-like wick structure 130i contacting the first casing 110i may have different thermal resistances. An inner wick structure 140i of the second casing 120i is included, also having different and corresponding thermal resistances to the sheet-like wick structure 130i. In some embodiments, the thermal resistances of the first part A1a, second part A1b, second portion A2, and third portion A3 of the second casing 120k are similar to the thermal resistances of the first part B1a, second part B1b, second portion B2, and third portion B3 of the sheet-like wick structure 130k, respectively. The thermal resistances of the first part A1a, second part A1b, second portion A2, and third portion A3 and first part B1a, second part B1b, second portion B2, and third portion B3 can be varied as required by application and design. A mounting area 184 includes a thickness and the mounting area 184 has a lower thermal resistance than that of the second part B1b, first part B1a, second portion B2, and third portion B3, respectively. A high temperature heat source, such as a processor (CPU, GPU, and the like) may be attached to the outer surface of the mounting area 184 of the first casing 110i. The thermal resistance of the second part A1b is lower than the first part A1a, and the thermal resistance of the first part A1a is lower than the thermal resistances of the second portion A2 and third portion A3, respectively. An additional heat source, such as a CPU may be attached to the outer surface of the second casing 120i opposite the first part A1a.

[0128] The wick structures 141k and 142k, portions of different thermal resistances, and the copper sintered powered wick layers surrounding the supporting structures 122, further improves circulation of cooling fluid in the vapor chamber 100i, for example, against gravity. Thus, the efficiency of the vapor chamber in dissipating heat is further improved.

[0129] In the embodiments above, the first part A1a and second part A1b of the inner wick structure 140i and corresponding first part B1a and second part B1b of the sheet-like wick structure 130i are disclosed as separate components positioned adjacent side by side each other. However, in

other embodiments, a wick structure is arranged within another wick structure.

[0130] FIG. 15 illustrates a plan view of a second casing **120i** including two wick structures, according to embodiments of the disclosure. FIG. 16 illustrates a plan view of a first casing including a sheet-like wick structure according to embodiments of the disclosure. The second casing **120i** having an inner wick structure **140i** may be used in the vapor chamber **100i** that includes a first casing **110i** and a sheet-like wick structure **130i**.

[0131] As illustrated in FIG. 15, the second casing **120i** having the inner wick structure **140i**, is generally wing-shape including a first part **A1c**, a second part **A1d** positioned within the first part **A1c**, a second portion **A2**, and a third portion **A3**, wherein the first part **A1c** and second part **A1d** are positioned between the second portion **A2** and third portion **A3**. The first part **A1c** has a width **W1**, the second part **A1d** has a width **W3**, and the second portion **A2** and the third portion **A3** each have a width **W2**. In some embodiments, the widths **W2** of the second portion **A2** and the third portion **A3** are about the same, and the widths **W2**, **W3** of the first part **A1c** and second part **A1d** are different. In some embodiments, the widths **W2** of the second portion **A2** and the third portion **A3** are smaller than the widths **W1**, **W3** of the first part **A1c** and second part **A1d**. In other embodiments, the second portion **A2** and the third portion **A3** have different widths and the first part **A1c** and second part **A1d** may have different widths, but the widths **W2** of the second portion **A2** and the third portion **A3** are smaller than the widths **W1**, **W3** of the first part **A1c** and second part **A1d**. As illustrated, the first part **A1c** completely surrounds the second part **A1d**. In other words, the second part **A1d** is located entirely within the first part **A1c**.

[0132] As illustrated in FIG. 16, the sheet-like wick structure **130i** is generally wing-shaped including a first part **B1c**, a second part **B1d** within the first part **B1c**, a second portion **B2**, and a third portion **B3**, wherein the first part **B1c** and second part **B1d** are positioned between the second portion **B2** and third portion **B3**. As illustrated, the first part **B1c** completely surrounds the second part **B1d**. In other words, the second part **B1d** is located entirely within the first part **B1c**. In some embodiments, the widths of the second portion **B2** and the third portion **B3** are about the same and the widths of the first part **B1c** and second part **B1d** are different. In some embodiments, the widths of the second portion **B2** and the third portion **B3** are smaller than the width of the second part **B1d**, and the width of the second part **B1** is smaller than the width of the first part **B1c**. In other embodiments, the second portion **B2** and the third portion **B3** have different widths, and the first part **B1c** and second part **B1d** have different widths, but the widths of the second portion **B2** and the third portion **B3** are smaller than the widths of the first part **B1c** and second part **B1d**.

[0133] In some embodiments, the thermal resistances of the second portion **B2** and the third portion **B3** are about the same, and the thermal resistances of the first part **B1c** and second part **B1d** are different. In some embodiments, the thermal resistances of the second portion **B2** and the third portion **B3** are higher than the thermal resistances of the first part **B1c** and second part **B1d**. In other embodiments, the second portion **B2** and the third portion **B3** have different thermal resistances, and the first part **B1c** and second part **B1d** have different thermal resistances, but the thermal resistances of the second portion **B2** and the third portion **B3** are greater than the thermal resistances of the first part **B1c** and second part **B1d**. Thus, the heat transfer performance of the first part **B1c** and second part **B1d** is higher than the heat transfer performance of the second portion **B2** and the third portion **B3**. In other embodiments, the thermal resistances of the second portion **B2** and the third portion **B3** are about the same, the thermal resistance of the first part **B1c** is less than the thermal resistances of the second portion **B2** and the third portion **B3** and the thermal resistance of the second part **B1d** is less than the thermal resistance of the first part **B1c**. In still other embodiments, the thermal resistance of the second part **B1d** is higher than the thermal resistance of the mounting area **184**. Thus, the second part **B1d** in thermal contact with the first heat source **H1** attached to the outer surface of the mounting area **184** dissipates heat more efficiently, and can be used to dissipate heat generated by high temperature heat sources, such as, central processing units (CPUs), graphic processing units (GPUs), and the like.

[0134] Referring to FIG. 15, the second heat source H2 is attached to the outer surface of the second casing **120i** and thereby the outer surface of the vapor chamber **100i** using a thermal paste, or other techniques. In some embodiments, the thermal resistances of the second portion A2 and the third portion A3 are about the same, and the thermal resistances of the first part A1c and second part A1d are different. In some embodiments, the thermal resistances of the second portion A2 and the third portion A3 are higher than the thermal resistances of the first part A1c and second part A1d. In other embodiments, the second portion A2 and the third portion A3 have different thermal resistances, and the first part A1c and second part A1d have different thermal resistances, but the thermal resistances of the second portion A2 and the third portion A3 are higher than the thermal resistances of the first part A1c and second part A1d. Thus, the heat transfer performance of the first part A1c and second part A1d is higher than the heat transfer performance of the second portion A2 and the third portion A3. In other embodiments, the thermal resistances of the second portion A2 and the third portion A3 are about the same, the thermal resistance of the first part A1c is lesser than the thermal resistances of the second portion A2 and the third portion A3, and the thermal resistance of the second part A1d is lesser than the thermal resistance of the first part A1c. Thus, the second part A1d in thermal contact with the second heat source H2 attached to the outer surface of the second casing **120i** dissipates heat more efficiently, and can be used to dissipate heat generated by high temperature heat sources such as central processing units (CPUs), graphical processing units (GPUs), and the like.

[0135] In some embodiments, the heat transfer performance of the second part B1d in thermal contact with the first heat source H1 attached to the mounting area **184** is about the same as the heat transfer performance of the second part A1d in thermal contact with the second heat source H2 attached to the outer surface of the second casing **120i**. In some embodiments, the heat transfer performance of the second part B1d in thermal contact with the first heat source H1 attached to the mounting area **184** is higher than the heat transfer performance of the second part A1d in thermal contact with the second heat source H2 attached to the outer surface of the second casing **120i**.

[0136] In some embodiments, the sheet-like wick structure contacting the first casing and inner wick structure of the second casing are copper mesh wick structures, or other types of wick structures. In some embodiments the thermal resistances of the first part A1c, second part A1d, second portion A2, and third portion A3 are similar to the thermal resistances of the first part B1c, second part B1d, second portion B2, and third portion B3, respectively. In some embodiments, the thermal resistances of the first part A1c, second part A1d, second portion A2, and third portion A3 are about the same as the thermal resistances of the first part B1c, second part B1d, second portion B2, and third portion B3, respectively. However, in other embodiments, the thermal resistances of the first part A1c, second part A1d, second portion A2, and third portion A3 are different from the thermal resistances of the first part B1c, second part B1d, second portion B2, and third portion B3, respectively. The thermal resistances can have any desired value as required by application or design provided, the heat transfer performance of the second part B1d in thermal contact with the first heat source H1 attached to the outer surface of the mounting area is relatively higher than or equal to the heat transfer performance of the second part in thermal contact with the second heat source H2 attached to the outer surface of the second casing.

[0137] The wick structures **141k** and **142k**, each, contact the sheet-like wick structure **130i** and the inner wick structure **140i** of the second casing **120i**. The wick structures **141k** and **142k** decrease a distance the cooling fluid has to circulate in the vapor chamber during operation of the vapor chamber **100i**.

[0138] When the vapor chamber is used in a vertical manner (e.g., in a position wherein the first heat source H1 and/or second heat source H2 are (is) located at a level higher than the second portion A2 and third portion A3), the cooling fluid in the vapor chamber evaporates and turns to vapor while absorbing the heat generated by the first and/or second heat sources H1 and/or H2, respectively. The thermal resistance of the second-part A1d is lower than the thermal resistance of

the first part **A1c**. The thermal resistance of the first part **A1c** is lower than the thermal resistances of the second portion **A2** and third portion **A3**. The heat transfer performance of the second part **B1d** in thermal contact with the first heat source **H1** is similar to the heat transfer performance of the second part **A1d** in thermal contact with the second heat source **H2**.

[0139] In some embodiments, the thermal resistance of the second part **B1d** is greater than the thermal resistance of the mounting area **184**. In some embodiments, the supporting structures **122** within the first part **A1c** and second part **A1d** are surrounded with the copper sintered powdered wick layer **124**, and this reduces the thermal resistance of the first part **A1c** and second part **A1d** having the first heat source **H1** and second heat source **H2**. As a result, heat transfer performance is increased. The cooling fluid (in vapor state) flows toward the second portion **A2** and the third portion **A3** and turns returns to liquid state, and the cooling fluid can flow to the second part **A1d** via the wick structures **141k** and **142k**. This configuration of the wick structures **141k** and **142k**, the portions of different thermal resistances, and the copper sintered powered wick layers surrounding the supporting structures decreases a temperature difference between the second part **A1d** and the second portion **A2** and the third portion **A3** by about 15 degrees Celsius or higher. The cooling fluid continuously circulates between the second part **A1d**, first part **A1c** and second portion **A2** and third portion **A3**, thereby dissipating the heat generated by the first and/or second heat sources **H1** and/or **H2**, respectively.

[0140] The wick structures **141k** and **142k** each have a flowing, generally sloping-ramped-shape and do not contact each other. Portions of the sheet-like wick structure **130i** contacting the first casing **110i** may have different thermal resistances. An inner wick structure **140i** of the second casing **120i** also has different thermal resistance compared to the sheet-like wick structure **130i**. In some embodiments, the thermal resistances of the first part **A1c**, second part **A1d**, second portion **A2**, and third portion **A3** of the second casing **120k** are similar to the thermal resistances of the first part **B1c**, second part **B1d**, second portion **B2**, and third portion **B3** of the sheet-like wick structure **130i**, respectively. The thermal resistances of the first part **A1c**, second part **A1d**, second portion **A2**, and third portion **A3** and first part **B1c**, second part **B1d**, second portion **B2**, and third portion **B3** may be changed as required by application and design for obtaining a desired performance. A second heat source **H2** may be attached to the outer surface of the second casing **120i** in addition to a first heat source **H1** being attached to the outer surface of the first casing **110i**. A mounting area **184** includes a thickness and the mounting area **184** has a lower thermal resistance than that of the second part **B1d**, first part **B1c**, second portion **B2**, and third portion **B3**, respectively. A high temperature heat source, such as a central processing unit (CPU, graphical processing unit (GPU), and the like may be attached to the outer surface of the mounting area **184** of the first casing **110i**. The thermal resistance of the second part **A1d** is lower than the first part **A1c**, and the thermal resistance of the first part **A1c** is lower than the thermal resistances of the second portion **A2** and third portion **A3**, respectively. One or more high temperature heat sources, may be attached to the outer surface of the second casing **120i** opposite the second part **A1d**. The wick structure **142k** overlaps the second heat source **H2** and first heat source **H1**. The wick structures **141k** and **142k**, varied thermal resistance portions, and the copper sintered powered wick layers surrounding the supporting structures **122** improve circulation of cooling fluid in the vapor chamber **100i**. Thus, the efficiency of the vapor chamber in dissipating heat is further increased.

[0141] Instead of the two wick structure some embodiments include a single wick structure. FIG. **17** illustrates a plan view of the second casing **120i** including a wick structure **141m**, according to embodiments of the disclosure.

[0142] The wick structure **141m** contacts the sheet-like wick structure **130i** contacting the first casing **110i** and inner wick structure **140i** of the second casing **120i**. The wick structure **141m** is a generally bell shape structure that is flattened at the top and is located in the gaps between the supporting structures **122**.

[0143] The wick structure **141m** includes a first section **131** that is disposed in the first part **A1a**

and second part **A1b**. The wick structure **141m** includes a second section **112** disposed in the second portion **A2** and is located at a lower level (referring to the orientation in FIG. 17) than the first section **131**. The second section **112** is arranged relatively horizontal and portions of the first section **131** that are connected to the second section **112** are inclined (e.g., positioned at an angle greater than 0° and less than 90°) relative to the second section **112**. The first section **131** is connected to the second section **112** at the connection between the first part **A1a** and second portion **A2**. As illustrated, the wick structure **141m** generally follows the shape of the casing **120i**. In an example, the first section **131** meets the second section **112** at angles or degrees of curvatures greater than or less than 45° , and greater than 0° and 90° .

[0144] The wick structure **141m** includes a third section **104** disposed in the third portion **A3** and is located at a lower level (referring to the orientation in FIG. 17) than the first section **131**. The third section **104** is arranged relatively horizontal and portions of the first section **131** that are connected to the third section **104** are inclined (e.g., positioned at an angle greater than 0° and less than 90°) relative to the third section **104**. In some embodiments, the second section **112** and the third section **104** are located at a same horizontal level (e.g., collinear). In some other embodiments, the second section **112** and the third section **104** are vertically offset from each other. The first section **113** is connected to the third section **104** at the connection between the second part **A1b** and third portion **A3**. In an example, the first section **113** meets the third section **104** at angles or degrees of curvatures greater than or less than 45° , and greater than 0° and 90° . The first section **131** of the wick structure **141m** and the second section **104** of the wick structure **141m** form the opposite ends of the wick structure **141m** and do not contact each other.

[0145] The first section **131**, located in the first part **A1a** and second part **A1b**, overlaps the first heat source **H1** attached to the outer surface of the mounting area **184** of the first casing **110i** and the second heat source **H2** attached to the outer surface of the second casing **120i**. The second section **112** includes an end **S3** of the wick structure **141m** in the second portion **A2**. The third section **104** includes an end **S4** of the wick structure **141m** in the third portion **A3**. In an example, the first and second heat sources **H1**, **H2**, are attached to the outer surfaces of the mounting area **184** of the first casing **110i** and second casing **120i**, in the second part **A1b** and first part **A1a**, respectively. The thermal resistance of the second part **A1b** is lower than the thermal resistance of the first part **A1a**. The thermal resistance of the first part **A1a** is lower than the thermal resistances of the second portion **A2** and third portion **A3**. The heat transfer performance of the second part **B1b** in thermal contact with the first heat source **H1** is higher than the heat transfer performance of the second part **A1b** in thermal contact with the second heat source **H2**.

[0146] As illustrated, the end **S3** of the wick structure **141m** is located proximate (non-contacting) an edge **A32** of the second portion **A2** and the end **S4** of the wick structure **141m** is located proximate (non-contacting) an edge **A33** of the third portion **A3** opposite the edge **A32**. The second sections **112** and **114** are disposed proximate (non-contacting) an edge **A31** of the second casing **120i** that forms (or otherwise defines) the first part **A1a**, the second part **A1b**, the second portion **A2**, and the third portion **A3** of the second casing **120i**. The edge **A31** connects the edge **A32** and edge **A33**. In other embodiments, the end **S3** may contact the edge **A32** and the end **S4** may contact the edge **A33**.

[0147] As illustrated, the first section **131** of the wick structure **141m** is located closer to the top edge **A34** of the second casing **120i** (and thereby the top edge of the vapor chamber **100i**) than the edge **A31**. In other embodiments, the first section **131** of the wick structure **141m** is located closer to the edge **A31** than the edge **A34**. The placement of the first section **131** is not limited in any regard, and the first section **131** can be located anywhere in the first part **A1a** and second part **A1b** provided the first section **131** (and thereby the wick structure **141m**) overlaps the first heat source **H1** and second heat source **H2**.

[0148] It will be understood that the location of the mounting area **184** may be varied depending on the application and design requirements. The mounting area **184** can be located at any desired

location provided the first heat source H1 is overlapped by the wick structure **141m**.

[0149] The wick structure **141m** decreases a distance the cooling fluid has to circulate in the vapor chamber **100i**. The length of the wick structure **141m** is not limited to any particular length and may be adjusted according to application and design requirements.

[0150] When the vapor chamber is used in a vertical manner (e.g., in a position wherein the first heat source H1 and/or second heat source H2 are located at a level higher than the second portion A2 and third portion A3), the cooling fluid in the vapor chamber evaporates and turns to vapor while absorbing the heat generated by the first and/or second heat sources H1 and/or H2, respectively. The thermal resistance of the second part A1b is lower than the thermal resistance of the first part A1a. The thermal resistance of the first part A1a is lower than the thermal resistances of the second portion A2 and third portion A3. The heat transfer performance of the second part B1b in thermal contact with the first heat source H1 attached to the outer surface of the mounting area **184** is greater than the heat transfer performance of the second part A1b in thermal contact with the second heat source H2 attached to the outer surface of the second casing **120m**. In some embodiments, the thermal resistance of the second part B1b is higher than the thermal resistance of the mounting area **184**. In some embodiments, the supporting structures **122** are surrounded by the copper sintered powdered wick layer **124** in the first part A1a and second part A1b, and this further decreases the thermal resistance of the first part A1a and second part A1b, and thereby increases the heat transfer performance. The cooling fluid (in vapor state) flows toward the second portion A2 and the third portion A3 and turns back to liquid state, and the cooling fluid can flow to the first part A1a and second part A1b via the wick structure **141m**. This configuration of the wick structure **141m**, the portions having different thermal resistances, and the copper sintered powered wick layers surrounding the supporting structures, decreases a temperature difference between the first part A1a and second part A1b and the second portion A2 and the third portion A3 by around 15 degrees Celsius or more. The cooling fluid continuously circulates between the first part A1a and second part A1b and second portion A2 and third portion A3, thereby dissipating the heat generated by the first and/or second heat sources H1 and/or H2, respectively.

[0151] FIG. **18** illustrates a plan view of the second casing **120i** including the wick structure **141m**, according to embodiments of the disclosure. In this embodiment, the second casing **120i** includes the inner wick structure **140i** including the first part A1c, the second part A1d positioned within the first part A1c, the second portion A2, and the third portion A3. The first part A1c and second part A1d are positioned between the second portion A2 and third portion A3. The structure of the second casing **120i** including the wick structure **141m** is similar in some respects to the second casing **120i** illustrated in FIG. **15** including wick structures **141k** and **142k**, and will be understood with reference thereto where like reference numerals indicate like components and are not described again in detail.

[0152] The foregoing outlines features of several embodiments or examples so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments or examples introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A heat dissipation device, comprising: a first casing including: a first inner surface having a first wick structure, the first wick structure including: a first portion; a second portion; and a third portion, wherein the first portion is disposed between the second portion and third portion; a first

outer surface opposite the first inner surface; and a first mounting plate for receiving a first heat source disposed on the first outer surface and located overlapping the first portion of the first wick structure; a second casing coupled to the first casing, the second casing including: a second inner surface having a second wick structure, the second wick structure including: a first portion opposite the first portion of the first wick structure; a second portion opposite the second portion of the first wick structure; and a third portion opposite the third portion of the first wick structure, wherein the first portion of the second wick structure is disposed between the second portion and third portion of the second wick structure; and a second outer surface opposite the second inner surface and including a second mounting plate different from the first mounting plate for receiving a second heat source, the second mounting plate located overlapping the first portion of the second wick structure; a third wick structure having at least two bends and arranged between the first inner surface and the second inner surface and arranged in the first portions of the first wick structure and the second wick structure, the second portions of the first wick structure and the second wick structure, and the third portions of the first wick structure and the second wick structure, the third wick structure at least partially overlaps the second mounting plate and the first mounting plate; and a plurality of columns arranged between the first and second inner surfaces, wherein thermal resistances of the first portion of the first wick structure, second portion of the first wick structure and third portion of the first wick structure are same as thermal resistances of the first portion of the second wick structure, second portion, and third portion, respectively, the thermal resistance of the first portion of the first wick structure is less than the thermal resistances of the second portion and the third portion of the first wick structure, and the thermal resistance of the first portion of the second wick structure is less than the thermal resistances second portion and third portion of the second wick structure.

2. The heat dissipation device of claim 1, wherein a thermal resistance of the first mounting plate is less than the thermal resistances of the first portion of the first wick structure and the first portion of the second wick structure.

3. The heat dissipation device of claim 1, wherein the first portion of the second wick structure comprises a first part and a second part and the first portion of the first wick structure comprises a first part and a second part, the first part of the first wick structure is opposite the first part of the second wick structure, and the second part of the first wick structure is opposite the second part of the second wick structure.

4. The heat dissipation device of claim 3, wherein the first part and second part, each of the first wick structure are adjacent each other, the first part and second part, each of the second wick structure are adjacent each other, the second portion of the first wick structure is adjacent the first part of the first wick structure, the second portion of the second wick structure is adjacent the first part of the second wick structure, the third portion of the first wick structure is adjacent the second part of the first wick structure, the third portion of the second wick structure is adjacent the second part of the second wick structure, and a thermal resistance of the second part of the first wick structure is less than the thermal resistance of the first part of the first wick structure and a thermal resistance of second part of the second wick structure is less than the thermal resistance of the first part of the second wick structure and the second mounting plate is disposed overlapping the first part of the first wick structure and the first part of the second wick structure and the first mounting plate is disposed overlapping the second part of the first wick structure and second part of the second wick structure.

5. The heat dissipation device of claim 3, wherein the first part of the first wick structure is within the second part of the first wick structure, the first part of the second wick structure is within the second part of the second wick structure, the second portion of the first wick structure is adjacent the second part of the first wick structure, the second portion of the first wick structure is adjacent to the second part of the second wick structure, the third portion of the first wick structure is adjacent the second part of the first wick structure, the third portion of the second wick structure is

adjacent the second part of the second wick structure, and wherein a thermal resistance of the first part of the second wick structure are less than a thermal resistance of the first part of the first wick structure, a thermal resistance of the second part of the first wick structure is less than a thermal resistance of the second part of the first wick structure, and the second mounting area and first mounting area are disposed overlapping the first part of the second wick structure and the first part of the first wick structure, respectively.

6. The heat dissipation device of claim 5, wherein a cross-sectional area of the first part of the second wick structure is greater than a cross-sectional area of the second part of the second wick structure, and a cross-sectional area of the second part of the second wick structure is greater than cross-sectional areas of the second portion and third portion of the second wick structure.

7. The heat dissipation device of claim 1, wherein each of the plurality of columns adjacent to the third wick structure are surrounded by porous wick structures.

8. The heat dissipation device of claim 1, wherein a cross-sectional area of the first portion of the first wick structure is greater than cross-sectional areas of the second portion and the third portion of the first wick structure.

9. The heat dissipation device of claim 1, wherein each of the plurality of columns overlapping the second mounting plate and first mounting plate are surrounded by porous wick structures.
